

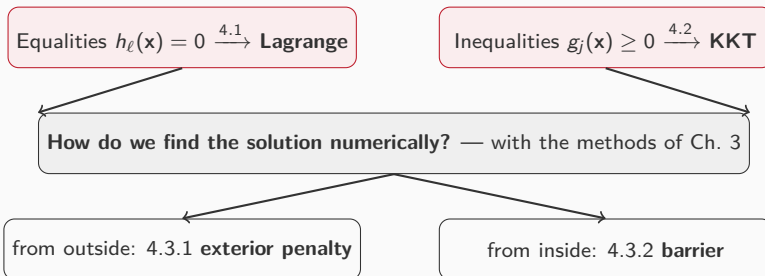
Optimization

4.1 Constrained extrema: Lagrange multipliers

José Firmino Aguilar Madeira

Chapter 4. Constrained optimization — map

When not all directions are allowed.



Ch. 3: choose a direction along which f decreases.

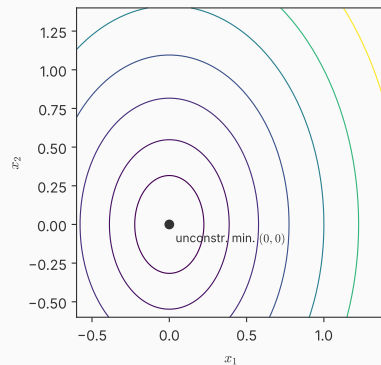
Ch. 4: that direction must also **respect what is allowed**.

CHARACTERIZE → 4.1–4.2: *what* distinguishes a solution (and the worth of the constraints: multipliers).

COMPUTE → 4.3: *how* to find it, turning the problem into a sequence of unconstrained problems.

What has changed since Chapter 3?

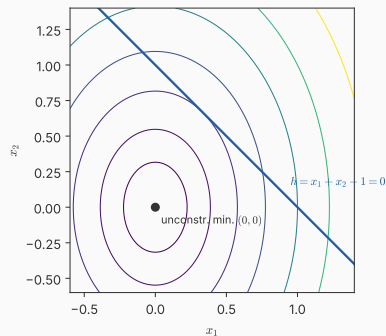
- **Ch. 3:** from $\mathbf{x}$ we could move in **any direction**;
at the minimum, $\nabla f(\mathbf{x}^*) = \mathbf{0}$.



$$\min 2x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1$$

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- **Ch. 3:** from $\mathbf{x}$ we could move in **any direction**; at the minimum, $\nabla f(\mathbf{x}^*) = \mathbf{0}$.
- **Now:** we can only move **along the constraint** $h(\mathbf{x}) = 0$ — here, the blue line.
- The unconstrained minimum $(0, 0)$ **is not even feasible**: it is not on the line.



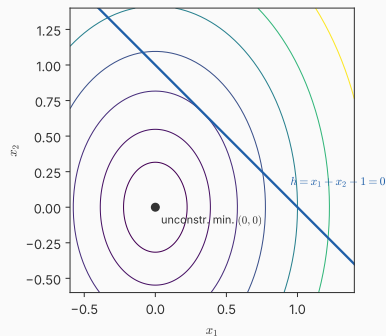
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Question

What replaces $\nabla f = \mathbf{0}$ when we can only move along $h(\mathbf{x}) = 0$?



$$\min 2x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 = 1$$

The general problem with equality constraints

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{s.t.} \quad h_\ell(\mathbf{x}) = 0, \quad \ell = 1, \dots, K \quad (K < n); \quad X = \{\mathbf{x} : h_\ell(\mathbf{x}) = 0\}.$$

- Sometimes we can eliminate a variable (the can from 1.1: $h = V/(\pi r^2)$; here h is the height) and return to Chs. 2–3. In general we **cannot** solve $h_\ell(\mathbf{x}) = 0$ explicitly.
- We need a condition written with f and h_ℓ as they are — without eliminating variables.
- The conditions of 3.1 ($\nabla f = \mathbf{0}$) no longer serve: the unconstrained minimum is generally not in X .

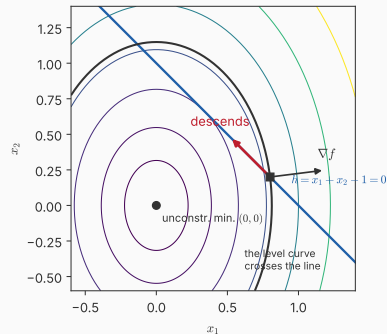
In the background: if the ∇h_ℓ are linearly independent, X is locally a “surface” (manifold) of dimension $n - K$ — a curve if $n = 2$, $K = 1$ (the example) or $n = 3$, $K = 2$.

Notation: from here on g_j (and $\mathbf{g}$) are constraints (4.2); the gradient is always written ∇f .

Geometry (1): while the line is being crossed, we can still descend

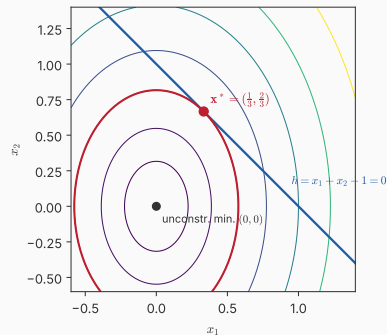
- At the point $\mathbf{x} = (0.8, 0.2)$ on the line, the level curve of f through it **crosses** the line.
- $\nabla f(\mathbf{x}) = (3.2, 0.4)$ has a component **along the line**. With the direction of the line $\mathbf{d} = (1, -1)$: $\nabla f(\mathbf{x})^T \mathbf{d} = 2.8 \neq 0$.
- Moving along the line in the direction $-\mathbf{d}$ (red arrow), f decreases: this point is **not** the minimum on the line.

As long as the level curve crosses the constraint, there is a feasible way to descend.



Geometry (2): at the optimum there is tangency

- Descending along the line we reach $\mathbf{x}^* = (\frac{1}{3}, \frac{2}{3})$: the level curve $f = \frac{2}{3}$ **touches** the line without crossing it — it is **tangent**.



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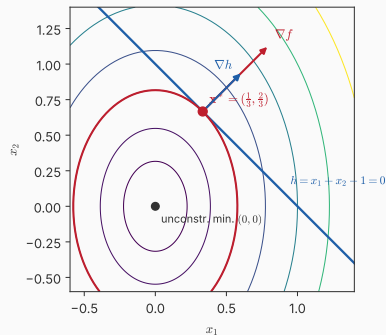
- Descending along the line we reach $\mathbf{x}^* = (\frac{1}{3}, \frac{2}{3})$: the level curve $f = \frac{2}{3}$ **touches** the line without crossing it — it is **tangent**.
- There ∇f is perpendicular to the line, just like ∇h : the two gradients are **parallel**.

$$\nabla f(\mathbf{x}^*) + \lambda \nabla h(\mathbf{x}^*) = \mathbf{0}$$

$$\left(\frac{4}{3}, \frac{4}{3}\right) + \lambda(1, 1) = \mathbf{0} \Rightarrow \lambda = -\frac{4}{3}.$$

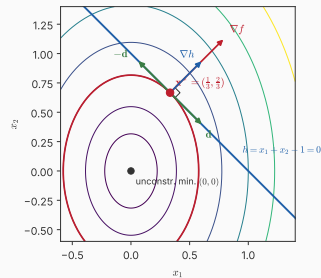
It is no longer $\nabla f = \mathbf{0}$: it is “ ∇f with no component along the constraint”.

There is no need to memorize the Lagrange condition: it can be read off the figure.



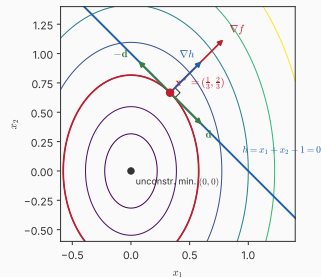
From tangency to the Lagrange condition

1. **d** **tangent** to the constraint (moving on it, 1st order):
 $\nabla h(\mathbf{x}^*)^T \mathbf{d} = 0$. Here $\mathbf{d} = (1, -1)$, $\nabla h = (1, 1)$.



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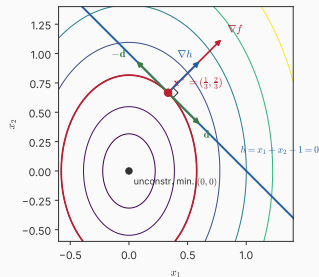
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2. At the minimum, f cannot vary to 1st order along $\mathbf{d}$ or $-\mathbf{d}$: $\nabla f(\mathbf{x}^*)^T \mathbf{d} = 0$.
(If $\neq 0$, one of $\pm \mathbf{d}$ would descend — see “Geometry (1)”.)



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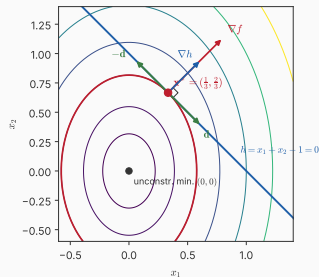
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With K constraints: $T = \{\mathbf{d} : \nabla h_\ell(\mathbf{x}^*)^T \mathbf{d} = 0, \ell = 1, \dots, K\}$, $T^\perp = \text{span}\{\nabla h_1(\mathbf{x}^*), \dots, \nabla h_K(\mathbf{x}^*)\}$;

hence $\nabla f(\mathbf{x}^*) + \sum_\ell \lambda_\ell \nabla h_\ell(\mathbf{x}^*) = \mathbf{0}$.

(If the $\nabla h_\ell(\mathbf{x}^*)$ are linearly independent, each $\mathbf{d} \in T$ is the direction of a curve staying in X — here the assumption enters.)



From geometry to the Lagrangian

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↓ computational form	$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{\ell} \lambda_{\ell} h_{\ell}(\mathbf{x})$ $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0} \text{ (} n \text{ eq.)}, \quad \nabla_{\boldsymbol{\lambda}} \mathcal{L} = \mathbf{0} \iff h_{\ell} = 0 \text{ (} K \text{ eq.)}$

The Lagrangian is not a new object: it is a convenient way of writing what geometry has already given
— $n + K$ equations, $n + K$ unknowns $(\mathbf{x}, \boldsymbol{\lambda})$.

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First-order necessary condition (Lagrange)

If $\mathbf{x}^*$ is a local minimum of f s.t. $h_{\ell}(\mathbf{x}) = 0$, with f, h_{ℓ} of class C^1 near $\mathbf{x}^*$, and $\nabla h_1(\mathbf{x}^*), \dots, \nabla h_K(\mathbf{x}^*)$ are **linearly independent**, then there exist $\lambda_1, \dots, \lambda_K$ (**Lagrange multipliers**) with $\nabla f(\mathbf{x}^*) + \sum_{\ell} \lambda_{\ell} \nabla h_{\ell}(\mathbf{x}^*) = \mathbf{0}$ and $h_{\ell}(\mathbf{x}^*) = 0$. The λ_{ℓ} are **free in sign**; the sign depends on the $\pm$ convention in $\mathcal{L}$ (the one used in class: $\mathcal{L} = f + \sum \lambda_{\ell} h_{\ell}$; with $f - \sum \lambda_{\ell} h_{\ell}$, as in Nocedal and Wright, only the sign of λ_{ℓ} changes).

The worked example

Recipe

1. Build $\mathcal{L}$. 2. Solve $\nabla_{\mathbf{x}}\mathcal{L} = \mathbf{0}$ and $h_{\ell} = 0$. 3. Check / classify the candidates.

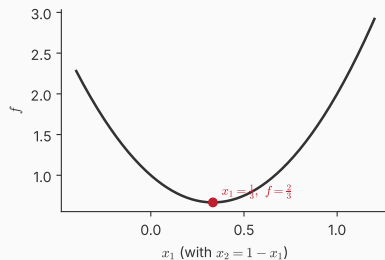
$$\mathcal{L} = 2x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1):$$

$$\partial_{x_1}\mathcal{L} = 4x_1 + \lambda = 0, \quad \partial_{x_2}\mathcal{L} = 2x_2 + \lambda = 0, \quad x_1 + x_2 = 1.$$

From the first two, $x_1 = -\lambda/4$, $x_2 = -\lambda/2$; in the third, $-\frac{3}{4}\lambda = 1$:

$$\mathbf{x}^* = \left(\frac{1}{3}, \frac{2}{3}\right), \quad \lambda^* = -\frac{4}{3}, \quad f^* = \frac{2}{3}$$

Lagrange only gives the candidate. Step 3: f restricted to the line, $\tilde{f}(x_1) = 2x_1^2 + (1 - x_1)^2$, is a convex parabola — it is a minimum, and global.



Lagrange gives candidates, not necessarily minima

Necessary $\neq$ sufficient

$\mathbf{x}^*$ local minimum (and ∇h_ℓ lin. indep.) $\implies$ Lagrange conditions

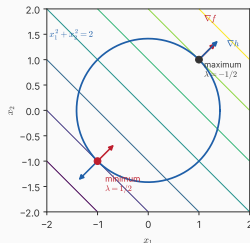
Lagrange conditions $\not\Rightarrow \mathbf{x}^*$ minimum

$\min x_1 + x_2$ s.t. $x_1^2 + x_2^2 = 2$:

$(1, 1) + \lambda(2x_1, 2x_2) = \mathbf{0} \Rightarrow x_1 = x_2 = -\frac{1}{2\lambda}$, and on the constraint $x_1 = x_2 = \pm 1$.

- $(-1, -1)$, $\lambda = \frac{1}{2}$, $f = -2$: **minimum**.
- $(1, 1)$, $\lambda = -\frac{1}{2}$, $f = 2$: **maximum** — and it satisfies the same conditions.

Lagrange produces candidates; they must then be classified. In 4.2 (KKT) it will be the same.



Classify: compare f at the candidates (the circle is compact and $\nabla h \neq \mathbf{0}$ on it, so the minimum exists and is among them), or second order: $\mathbf{d}^T \nabla_{\mathbf{x}}^2 \mathcal{L} \mathbf{d} > 0$ for $\mathbf{d} \in T$, $\mathbf{d} \neq \mathbf{0}$; here $\nabla_{\mathbf{x}}^2 \mathcal{L} = 2\lambda I$: positive definite at $(-1, -1)$, negative at $(1, 1)$. With $n = 2$ and one constraint: bordered Hessian (next).

Classifying: the bordered Hessian ($n = 2$, one constraint)

At a candidate $(\mathbf{x}^*, \lambda^*)$, with $h_{x_i} = \partial h / \partial x_i$
and $\mathcal{L}_{ij} = \partial^2 \mathcal{L} / \partial x_i \partial x_j$:

$$\bar{H} = \begin{bmatrix} 0 & h_{x_1} & h_{x_2} \\ h_{x_1} & \mathcal{L}_{11} & \mathcal{L}_{12} \\ h_{x_2} & \mathcal{L}_{21} & \mathcal{L}_{22} \end{bmatrix}$$

Worked example, $(\frac{1}{3}, \frac{2}{3})$,
 $\nabla_{\mathbf{x}}^2 \mathcal{L} = \text{diag}(4, 2)$:

$$\det \begin{bmatrix} 0 & 1 & 1 \\ 1 & 4 & 0 \\ 1 & 0 & 2 \end{bmatrix} = -6 < 0$$

minimum

Circle, $(-1, -1)$, $\lambda = \frac{1}{2}$, $\nabla_{\mathbf{x}}^2 \mathcal{L} = I$: Circle, $(1, 1)$, $\lambda = -\frac{1}{2}$, $\nabla_{\mathbf{x}}^2 \mathcal{L} = -I$:

$$\det \begin{bmatrix} 0 & -2 & -2 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} = -8 < 0$$

minimum

$$\det \begin{bmatrix} 0 & 2 & 2 \\ 2 & -1 & 0 \\ 2 & 0 & -1 \end{bmatrix} = 8 > 0$$

maximum

Rule

$\det \bar{H} < 0 \Rightarrow$ local minimum; $\det \bar{H} > 0 \Rightarrow$ local maximum; $\det \bar{H} = 0$: no conclusion.

Why. Expanding the determinant, $\det \bar{H} = -\mathbf{d}^T \nabla_{\mathbf{x}}^2 \mathcal{L} \mathbf{d}$ with $\mathbf{d} = (h_{x_2}, -h_{x_1})$, tangent to the constraint ($\nabla h^T \mathbf{d} = 0$). So $\det \bar{H} < 0$ is the second-order condition $\mathbf{d}^T \nabla_{\mathbf{x}}^2 \mathcal{L} \mathbf{d} > 0$ for $\mathbf{d} \in T$ (Nocedal and Wright, 2006, Thm. 12.6).

With more variables or constraints one uses minors of $\bar{H}$; in this course $n = 2$, $K = 1$ is enough.

What does λ mean?

Change the constraint to $x_1 + x_2 = 1 + \delta$. The same calculations give

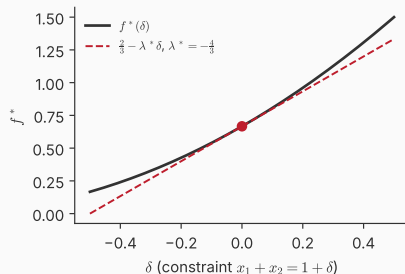
$$\mathbf{x}^*(\delta) = (1 + \delta)\left(\frac{1}{3}, \frac{2}{3}\right), \quad f^*(\delta) = \frac{2}{3}(1 + \delta)^2 \approx \frac{2}{3} + \frac{4}{3}\delta.$$

The slope at $\delta = 0$ is $\frac{4}{3} = -\lambda^*$: $\lambda^* = -\frac{df^*}{d\delta}$.

λ^* measures (up to sign) the sensitivity of the optimal value to a small change in the constraint (in general, under regularity, with $h_\ell(\mathbf{x}) = a_\ell(\mathbf{x}) - c_\ell$: $\lambda_\ell = -\partial f^* / \partial c_\ell$).

Remember 1.1?

In the production plan, “an extra hour of machine time is worth 10 €”. This is it: how much the optimum changes per extra unit of resource. (There the constraints are inequalities — it is the shadow price of 4.2.)



Example 2: the can from 1.1, without eliminating variables

$\min A(r, h) = 2\pi r^2 + 2\pi rh$ s.t. $\pi r^2 h = V$ ($V = 330 \text{ cm}^3$; here h is the height). $\mathcal{L} = 2\pi r^2 + 2\pi rh + \lambda(\pi r^2 h - V)$:

$$\partial_r \mathcal{L} = 4\pi r + 2\pi h + 2\lambda\pi rh = 0$$

$$\partial_h \mathcal{L} = 2\pi r + \lambda\pi r^2 = 0 \Rightarrow \lambda = -2/r$$

$$\partial_\lambda \mathcal{L} = \pi r^2 h - V = 0$$

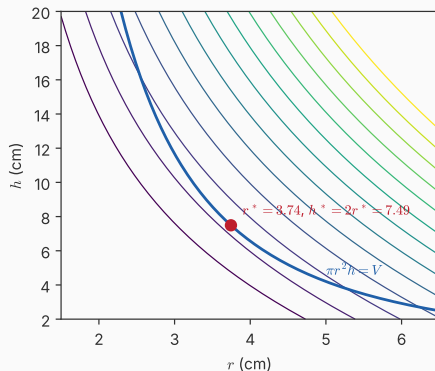
Substituting $\lambda = -2/r$ in the first:

$4\pi r + 2\pi h = 4\pi h \Rightarrow h = 2r$ (height equal to diameter — without having solved the constraint for h).

$$r^* = \left(\frac{V}{2\pi}\right)^{1/3} = 3.74 \text{ cm}, \quad h^* = 7.49 \text{ cm}, \quad A^* = 264.4 \text{ cm}^2,$$

r^*, h^* as in 1.1. $\lambda^* = -2/r^* = -0.534 \text{ cm}^2/\text{cm}^3$;

$dA^*/dV = -\lambda^*$: each extra cm^3 of volume costs $\approx 0.53 \text{ cm}^2$ of sheet metal.



Several equality constraints

$$\min x_1^2 + x_2^2 + x_3^2 \quad \text{s.t.} \quad h_1 = x_1 + x_2 + x_3 - 3 = 0, \quad h_2 = x_1 - x_2 - 1 = 0.$$

$n = 3$, $K = 2$: X is the line where the two planes meet — we seek its point closest to the origin.

$$\mathcal{L} = f + \lambda_1 h_1 + \lambda_2 h_2, \quad \nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}:$$

$$2x_1 = -\lambda_1 - \lambda_2, \quad 2x_2 = -\lambda_1 + \lambda_2, \quad 2x_3 = -\lambda_1.$$

$$h_2: x_1 - x_2 = -\lambda_2 = 1; \quad h_1: -\frac{3}{2}\lambda_1 = 3 \Rightarrow \lambda_1 = -2.$$

$$\boxed{\mathbf{x}^* = \left(\frac{3}{2}, \frac{1}{2}, 1\right), \quad \boldsymbol{\lambda}^* = (-2, -1), \quad f^* = \frac{7}{2}}$$

The same geometry

- Direction of the line: $\mathbf{d} = (1, 1, -2)$, with $\nabla h_1^T \mathbf{d} = \nabla h_2^T \mathbf{d} = 0$.
- $\nabla f(\mathbf{x}^*) = (3, 1, 2)$: $\nabla f^T \mathbf{d} = 3 + 1 - 4 = 0$ — no tangent component.
- $\nabla f = 2(1, 1, 1) + 1(1, -1, 0) = -\lambda_1^* \nabla h_1 - \lambda_2^* \nabla h_2$: in the span of the normals.
- f along the line is a convex parabola: global minimum.

When the Lagrange condition fails

$$\min x_1 + x_2 + x_3^2 \quad \text{s.t.} \quad h_1 = x_1 - 1 = 0, \quad h_2 = x_1^2 + x_2^2 - 1 = 0.$$

- The two constraints together force $x_1 = 1$, $x_2 = 0$:
 $X = \{(1, 0, x_3)\}$; the minimum is $\mathbf{x}^* = (1, 0, 0)$, $f^* = 1$.
- At $\mathbf{x}^*$: $\nabla h_1 = (1, 0, 0)$, $\nabla h_2 = (2, 0, 0)$ — **linearly dependent**.
- $\nabla f = (1, 1, 0)$ has a component in x_2 that no $\lambda_1 \nabla h_1 + \lambda_2 \nabla h_2$ can cancel: $\partial \mathcal{L} / \partial x_2 = 1 \neq 0$ for all λ .

The Lagrange system **has no solution** and yet $\mathbf{x}^*$ is the minimum.

Why it fails

The “tangents” computed with the gradients,

$T = \{\mathbf{d} : \nabla h_\ell^T \mathbf{d} = 0\} = \{\mathbf{d} : d_1 = 0\}$,
include the x_2 direction; but in X one can only move in x_3 . The step “ $\mathbf{d}$ tangent $\Rightarrow \nabla f^T \mathbf{d} = 0$ ” no longer holds.

Independence of the ∇h_ℓ (constraint qualification) guarantees that T really are the directions in which one can move. It returns in 4.2 under the name LICQ.

KEY TAKEAWAYS — 4.1

1. With $h(\mathbf{x}) = 0$ only **tangent directions** to the constraint are allowed: $\nabla h_\ell(\mathbf{x})^T \mathbf{d} = 0$.
2. At a regular optimum (∇h_ℓ linearly independent), ∇f **has no tangential component**:

$$\nabla f(\mathbf{x}^*) + \sum_{\ell} \lambda_{\ell} \nabla h_{\ell}(\mathbf{x}^*) = \mathbf{0}.$$

3. The **Lagrangian** $\mathcal{L} = f + \sum_{\ell} \lambda_{\ell} h_{\ell}$ turns this condition into a system of $n + K$ equations: $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}, h_{\ell} = 0$.
4. The Lagrange conditions are **necessary, not sufficient**: they give candidates, to be classified ($n = 2$, one constraint: $\det \bar{H} < 0$ minimum, $\det \bar{H} > 0$ maximum).
5. λ_{ℓ} measures (up to sign) the **sensitivity** of f^* to constraint ℓ .

Next (4.2): **inequalities** $g_j(\mathbf{x}) \geq 0$ — the sign of the multiplier now matters ($\lambda_j \geq 0$) and only *active* ones count (KKT); with $\mathcal{L} = f - \sum \lambda_j g_j - \sum \beta_{\ell} h_{\ell}$, the equalities have $\beta_{\ell} = -\lambda_{\ell}$.

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 12.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987.

Resources: — (see 4.2, “On the computer”).